

TEST_SUITE_GUIDE

Test Suite Guide

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Reference for running, writing, and interpreting tests in this repo.

Directory layout

Directory	Purpose	Hermetic?
tests/unit/	Python unit tests; mock external services	yes
tests/unit/merge_service/	Merge-service unit tests (deduplicator, search, classifier)	yes
tests/contract/	API schema / response contract checks	mostly
tests/integration/	Cross-component tests against the running stack	no
tests/e2e/	Full-browser (Playwright) and multi-service scenarios	no
tests/concurrency/	Semaphore / fan-out concurrency guarantees	yes
tests/property/	Hypothesis property tests	yes
tests/memory/	Tracemalloc / leak checks	yes
tests/observability/	Prometheus + metrics scrapes	yes
tests/security/	Auth-bypass / CSRF protection	no
tests/stress/	High-concurrency stress for the merge service	no
tests/benchmark/	Wall-clock performance budgets	no

Directory	Purpose	Hermetic?
tests/ performance/	Similar to benchmark, latency-focused	

Running tests

Hermetic suites (fast, ~15 s total)

```
python3 -m pytest tests/unit tests/contract tests/concurrency
        tests/property \
            tests/memory tests/observability --import-
mode=importlib
```

These tests don't need any services running. Suitable for pre-commit and CI gate.

Live-service suites (slow, 30-90 min)

```
./start.sh -p          # start containers (idempotent)
python3 webui-bridge.py # separate shell or systemd user service
SERVICE_PROBE_TIMEOUT=15 SERVICE_PROBE_RETRIES=10 \
python3 -m pytest tests/integration tests/e2e --import-
mode=importlib \
        --timeout=300 -p no:randomly
```

The live-service fixtures in tests/fixtures/services.py error (not skip) when services are down — so a failing run here means the stack needs attention.

Stress / benchmark / security (opt-in)

```
python3 -m pytest tests/stress tests/benchmark tests/security
```

These drive the service hard — don't run them on a production host.

Test-writing rules

1. **No skips.** Every test must either pass or fail. If a test needs a specific service, use the matching `*_live` fixture (see tests/fixtures/services.py) — the fixture errors loudly when the service is down.
2. **No data-dependent conditionals.** Don't write `if not results: pytest.skip(...)`. Use a query guaranteed to return data ("linux"), assert on the result, or mock the upstream. The no-results branch hides breakage.

3. **Use `timeout=300`** on any blocking `/api/v1/search/sync` call — the fan-out can take 90-150 s under realistic load.
4. **Use `timeout=30`** on trivial GET endpoints (`/`, `/dashboard`, `/health`, `/api/v1/config`). They're fast on idle probes but can wait a few seconds when the orchestrator is busy.
5. **Use `POST /api/v1/search`** (non-blocking) if the test is about the async kick-off contract. Use `POST /api/v1/search/sync` if the test needs actual results.
6. **Generate unique magnet hashes per file** via `secrets.token_hex(20)`. qBittorrent rejects duplicate torrent adds with Fails. — a shared hash across tests breaks flaky.

Error classifications

When a tracker returns zero results, the merge service now tags the `TrackerSearchStat` with a structured error. See `docs/MERGE_SEARCH_DIAGNOSTICS.md` for the full classifier table — short version:

- `upstream_http_403 / _404 / _timeout` — CDN / DNS issues
- `dns_failure / tls_failure` — domain is down
- `plugin_env_missing / plugin_parse_failure / plugin_crashed` — plugin-side bug (open an issue + patch the plugin)
- `deadline_timeout` — plugin ran the full 25 s with no results
- `auth_failure / upstream_captcha` — private-tracker login wall

Tests that search “linux” and check `tracker_stats[].error_type` should cover at least these common cases.

Live service toggles (env vars)

Var	Default	Effect
<code>PUBLIC_TRACKER_DEADLINE_SECONDS</code>	25	per-plugin deadline, clamped [5, 120]
<code>MAX_CONCURRENT_TRACKERS</code>	5	per-fan-out parallel plugin limit
<code>MAX_CONCURRENT_SEARCHES</code>	8	orchestrator-level search concurrency
<code>ENABLE_DEAD_TRACKERS</code>	0	force dead trackers back in
<code>SERVICE_PROBE_TIMEOUT</code>	3	fixture probe timeout
<code>SERVICE_PROBE_RETRIES</code>	5	fixture probe retries

Continuous integration

CI is currently manual (`./ci.sh`). Recommended gate:

1. Hermetic suites — run every push (fast, no infra dependency).
2. Integration + e2e — run nightly or pre-release (needs live stack).
3. Stress — ad-hoc (before deploys that touch the orchestrator).